

Industry connect

This month, take a note of the words of wisdom from eminent personalities in the tech industry.

**Cancer Control**

IBM's Microfluidic Probe seems like a step closer to curing cancer. Read this article to know more: <http://digit.in/17atgc1>

SETTING UP YOUR OWN NAS

THE HARD WAY!

There are plenty of out-of-the-box NAS solutions available, but not all of them are economical for a home/small office use. Here's how you get around to building a full fledged NAS device and save a lot of money along the way.

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NAS (Network Attached Storage) devices are gaining in popularity by the day given the versatility that these devices have to offer. Not only do they offer a communal storage which is pretty much the basic function of these devices but they can be modified using the numerous plugins available to suit a lot of use case scenarios. You could always buy a NAS device with a similar level of versatility which should set you back by quite a bit. Or you could repurpose an old computer as a NAS device. We'll walk you through the entire process.



Benefits

While adding a new hard drive to your computer does take care of your space shortage issues it only lasts till someone else runs out of space. Another computer then needs to be upgraded. So instead of upgrading multiple devices one by one you could just build a NAS device and get a shared storage space which amounts to a much more efficient usage of space and funds. Not to mention that particular storage can be accessed anytime by anyone and is not dependent on any one user's machine. Sharing a folder on your computer does get the job done and others can store data on your machine.

However, when your machine is turned off or experiences connectivity issues then work might come to a standstill till your machine is fixed.

Adding additional storage space requires a fair bit of configuration too not to mention searching through all these files across multiple machines is quite inefficient. A NAS is safe from network issues affecting individual machines. All files are indexed by most file systems and thus searching for a certain file is faster and required no user action to manually index newer partitions/drives. Adding a new drive to a NAS device is also an effortless procedure as a lot of motherboards support hot-plugging on SATA ports. So you don't even have to turn off the NAS device to add extra storage, just plug in your power and SATA cables and the device's operating system does the rest.

You'd be surprised how economical it is to setup a NAS device, a lot of NAS operating systems are open source. Hardware requirements aren't outlandish, an old PC can be easily repurposed to be a NAS device. Newer network infrastructure isn't required unless your work is bandwidth exhaustive. And by virtue of being open source, any problems that you could face along the entire process has already been documented and steps to resolve them are available quite easily. And just like a good portable storage these things come with automatic backup facility to another

drive or to the cloud. Finally, depending upon which file system you've chosen you even have data redundancy so in the event one of your hard drives fails the device will send you an alert well before the actual event occurs and all you have to do is swap the faulty drive with a new one. The data copying and restoring is done automatically by the device.

Software needed

There are umpteen operating systems available for NAS devices. Most of them are based on a Linux kernel. We've chosen FreeBSD based FreeNAS for this little tutorial. There is also U-NAS, unRAID, OpenFiler, Greyhole, NexentaStor etc and for the more adventurous sort – any linux distro on the planet. FreeNAS is pretty much the most popular distro out there for NAS and here's why:

ZFS is a file system designed for enterprise storage solutions and is a bit heavy on resources but is something you just setup once and forget about it forever. It is faster than most file systems while ensuring data security all along.



FreeBSD-based FreeNAS